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$$f(x) = \frac{8x^2}{x^4 + 2x^3 - 2x - 1} = \frac{A}{(x+1)^3} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)} + \frac{D}{(x-1)}$$

$$\Rightarrow \begin{aligned} A &= \left. \frac{8x^2}{(x-1)} \right|_{x=-1} = -4 \\ D &= \left. \frac{8x^2}{(x+1)^3} \right|_{x=1} = 1 \end{aligned}$$

$$f(x) - \frac{-4}{(x+1)^3} = \frac{8x-4}{(x+1)^2(x-1)}$$

$$\Rightarrow B = \left. \frac{8x-4}{(x-1)} \right|_{x=-1} = 6$$

$$\frac{8x-4}{(x+1)^2(x-1)} - \frac{6}{(x+1)^2} = \frac{2}{(x+1)(x-1)}$$

$$\Rightarrow C = \left. \frac{2}{(x-1)} \right|_{x=-1} = -1$$

$$\begin{aligned} \Rightarrow \int \frac{8x^2}{x^4 + 2x^3 - 2x - 1} &= -4 \int \frac{1}{(x+1)^3} + 6 \int \frac{1}{(x+1)^2} - \int \frac{1}{(x+1)} + \\ &\int \frac{1}{(x-1)} \\ &= \frac{2}{(x+1)^2} - \frac{6}{(x+1)} - \ln|x+1| + \ln|x-1| + k \\ &= \frac{-6x-4}{(x+1)^2} + \ln \left| \frac{x-1}{x+1} \right| + k \end{aligned}$$

References